

BANK OF UGANDA

Innovation Management System Requirements

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BANK OF UGANDA PROPRIETARY

Use Pursuant to Bank Regulations

1 INTRODUCTION

1.1 Background

Given the ever-evolving operating environment, the Bank of Uganda (BOU) Strategic Plan (SP) 2022-2027 recognized the critical need to develop an Innovation Strategy that would foster a culture that enables creative thinking, experimentation, and adoption of new ideas. The Innovation Strategy was successfully vetted and approved by the Board in September 2025, positioning the Bank to proactively respond to emerging opportunities and challenges.

One of the key Initiatives in the Innovation Strategy is the development of an Innovation Management System (IMTS). Currently, the innovation management process relies on paper forms and manual processes that are susceptible to errors, inconsistencies, and delays. The IMTS shall remedy this by automating the process from idea submission through to implementation. It will also support monitoring, evaluation and reporting on the performance of the Innovation Strategy by providing reliable data on KPIs such as idea submission rates, as well as turnaround time for processing of innovation ideas.

1.2 Purpose of Document

This document outlines the requirements that need to be met for the Innovation Management System to enable efficient processing of innovation ideas.

1.3 Scope of the Document

The scope of the document is limited to the following:

- Innovation Management System Functional requirements.
- Innovation Management System Non-functional requirements.

1.4 Technical Specification Format

The specification presented in this document are organized in the form of tables as below:

Requirement ID	Requirement	Importance	Notes
IMTS-REQ-XXX	(Text describing the requirement)	(Scale of importance from highest 5 to lowest 0 or unknown Z)	(One or more clarifying notes)

Where:

- IMTS-REQ-XXX** is a label that fully uniquely identifies a given requirement.
- XXX** is a positive integer (with leading zeros).

1.5 Target Audience

The target audience includes:

- Innovation Management and Business Process Improvement Section in the Strategy and Innovations Department
- Business Automation Department
- Information Technology Operations Department

1.6 References

The following are the reference documents used in the preparation of this document:

- BoU Innovation Strategy 2025-2027
- BoU Innovation Management Policy and Guidelines

1.7 Definition of Abbreviations and Acronyms

Acronym / Abbreviation	Definition / Description
BAUD	Business Automation Department

Acronym / Abbreviation	Definition / Description
BOU	Bank of Uganda
CIS	Center for Internet Security
CSV	Comma-Separated Values
HOD	Head of Department
HTTPS	HyperText Transfer Protocol Secure
ICB	Innovation Concept Brief
IIS Server	Internet Information Services Server
IM	Innovation Management Technical Team
IMBPI	Innovation Management and Business Process Improvement
IMTS	Innovation Management System
IP	Internet Protocol
ITO	Information Technology Operations
MS SQL Server	Microsoft Structured Query Language Server
OS	Operating System
OWASP	Open Web Application Security Project
PDF	Portable Document Format
SID	Strategy and Innovation Department
SSL	Secure Sockets Layer

1.8 Organization of the Document

This document is organized into the following main sections:

- a. Introduction
- b. Current Business Process and Challenges
- c. Proposed System Architecture
- d. Requirement specifications
- e. Appendices

1.9 Warning on ownership and publication

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2 CURRENT BUSINESS PROCESS AND CHALLENGES

2.1 Current Process

The Innovation Management and Business Process Improvement (IMBPI) Section in the Strategy and Innovation Department is responsible for receiving innovation ideas from Staff and supporting selection, concept development, experimentation/research, as well as deployment of approved solutions.

The IMBPI Section selects ideas that qualify as innovations after which the innovators develop Innovation Concept Briefs (ICBs). The ICBs are evaluated by the Innovation Management Technical Team (IMTT) and recommendations are approved by the Governor or Deputy Governor. Approved innovations progress to the experimentation/research stage, while those that do not require experimentation/research progress directly to deployment. Innovations that pass the experimentation/research stage are approved by the Governor or Deputy Governor for deployment. The IMBPI conducts monitoring, evaluation and reporting on Innovation Performance throughout the innovation process.

2.2 Drivers for change

Reasons for the change:

- a. The current Innovation Management Process relies heavily on manual data recording,

communications, and performance tracking which is a labor-intensive and error-prone practice.

- b. It is not only time-consuming but also susceptible to data inconsistencies, misplaced records, and lack of real-time access. As a result, it hinders operational efficiency, data accuracy, and timely decision-making.

Benefits of the change:

There is need for a system that can address the challenges associated with manual Innovation Management to realize benefits including the following:

- a. Data consistency
- b. Real Time access of records
- c. Improved data accuracy
- d. Reduction of human errors
- e. Automated report generation
- f. Improved performance management
- g. Timely communication

3 REQUIREMENT SPECIFICATIONS

3.1 Functional Requirements

This section specifies the functions that the Innovation Management System shall be able to perform.

3.1.1 Staff User Functional Requirements

Requirement ID	Requirement	Description	Remarks
IMTS-REQ-001	Submission of ideas	The system shall enable users to submit an unlimited number of ideas as individuals or a group.	
IMTS-REQ-010	Innovation submission form	The system shall provide an online innovation submission form for Staff to fill with specific required information regarding the ideas they are submitting. This shall include a provision for attachments.	

Requirement ID	Requirement	Description	Remarks
IMTS-REQ-015	View and edit	The system shall enable users to view and amend their own ideas except when locked for review by the Innovation Team.	
IMTS-REQ-020	Cancel/retract innovation idea	The system shall enable canceling submission or retracting an innovation idea by the end user.	
IMTS-REQ-025	Track status	The system shall enable users to track the status of their ideas in real time on the system.	
IMTS-REQ-030	Email notifications	The system shall enable email notifications to staff concerning their innovation ideas, e.g. regarding comments and status changes.	
IMTS-REQ-035	System notifications	The system shall have a notifications tab where Staff are notified of key activity on their accounts e.g., comments on their ideas and status changes.	

3.1.2 Innovation Team Functional Requirements

Requirement ID	Requirement	Description	Remarks
IMTS-REQ-040	View	The system shall enable the Innovation Team to view all submitted ideas.	
IMTS-REQ-045	Sort and filter	The system shall enable the Innovation Team to sort and filter ideas by status, category, department, date, etc.	
IMTS-REQ-050	Manage categories	The system shall enable the Innovation Team to create, edit, and remove categories.	
IMTS-REQ-	Review	The system shall enable the Innovation Team	

Requirement ID	Requirement	Description	Remarks
055		to access all idea submitter information (name, email, department, rank), idea details as filled in the submission form, and attachments.	
IMTS-REQ-060	Change stage	The system shall enable the Innovation Team to change idea stage (submitted, concept development, experimentation/research, deployment, closed).	
IMTS-REQ-065	Change Status	The system shall enable the Innovation Team to change idea status (pending information, under review, approved, declined).	
IMTS-REQ-070	Communicate	The system shall enable the Innovation Team to communicate with idea submitters (via comments and system notifications) to provide feedback.	
IMTS-REQ-075	Set timelines	The system shall enable the Innovation Team to set timelines within which stages for idea development have to be completed by both Staff users and the Innovation Team as follows: a. Idea selection - 30 days b. Concept development - 60 days c. Experimentation - 90 days d. Launch after approval - 30 days	
IMTS-REQ-080	Set user permissions/roles	The system shall enable the key user (Deputy Director Innovation & Project Support - DDIPS) from the Innovation Team to change permissions/roles for other users e.g., user, reviewer/manager.	

3.1.3 Documentation Requirements

Requirement ID	Requirement	Description	Remarks
-	Upload of resources	The system shall enable the Innovation Team to add, amend, or remove key resources such as the Innovation Strategy, Innovation Policy and Guidelines, and other key documents.	
-	Access to resources	The system shall enable the user to access key resources such as the Innovation Strategy, Innovation Policy and Guidelines, and other related documents in downloadable formats.	

3.1.4 Reporting Requirements

Requirement ID	Requirement	Description	Remarks
-	Generate report	The system shall enable the Innovation Team to extract a report in common formats (PDF, Excel) that can enable tracking of KPIs. The report shall include idea title (as per the submission form), date of submission, date of status change, date of stage change, as well the name of the performer of each activity.	

3.1.5 IT Admin Functional Requirements

Requirement ID	Requirement	Description	Remarks
IMTS-REQ-080	Create user accounts	The system shall enable the IT Administrator to create user accounts with necessary details such as username and password.	
IMTS-REQ-085	Reset user passwords	The system shall enable an IT Administrator to change user passwords as needed to	

Requirement ID	Requirement	Description	Remarks
		enhance security.	
IMTS-REQ-090	Lock user accounts	The system shall enable an IT Admin to lock user accounts in cases of unauthorized access to secure user data.	
IMTS-REQ-095	Unlock user accounts	The system shall enable an IT Admin to unlock user accounts.	

3.2 Non-Functional Requirements

The purpose of this section is to describe the non-functional requirements that must be efficiently met by the Proposed System Architecture, design, Security, reliability, data integrity, business processes and operations of the IMTS.

3.2.1 Proposed System Architecture

The Innovation Management System shall be a web application with both frontend and backend components that will store information in the database as described below:

- The web application shall only be accessed within the Bank's network using a web browser on the user's workstation.
- The system shall be shared from the IIS Server.
- The system shall store logs and audit details on the MS SQL database server.
- The system shall read data from the MS SQL database server.
- The system shall use BOU mail server to send notifications to recipients.

3.2.2 General requirements

Requirement ID	Requirement	Description	Remarks
IMTS-REQ-100	User friendly	The user interface must be intuitive and allow a first-time user to submit an idea with minimal training (usability).	

Requirement ID	Requirement	Description	Remarks
IMTS-REQ-105	Response time	Once the required information is filled, the system should execute any command in under 5 seconds under normal load and proper functioning of any external infrastructure dependencies.	
IMTS-REQ-110	System availability	The system shall achieve a minimum of 99.5% uptime measured over each quarter considering proper functioning of any external infrastructure dependencies and excluding pre-communicated maintenance windows.	
IMTS-REQ-115	Concurrent users	The system shall allow concurrent activity of at least 50 simultaneous users with no degradation in performance of more than 10%.	
IMTS-REQ-120	Data consistency	The system shall enable data consistency in multi-reviewer environments (concurrency control) so two reviewers cannot simultaneously override the same idea without detecting and resolving conflict.	
IMTS-REQ-125	Audit trail	The system shall log all user actions including log ins, submissions, and edits to enable traceability.	
IMTS-REQ-130	Back up	The data on the system shall be regularly backed up to ensure recovery and continuity.	
IMTS-REQ-135	Automated reminders	The system shall enable automated email reminders to both users and the Innovation Team to complete specific tasks within preset timelines.	
IMTS-REQ-140	Compatibility with web browsers	The system shall be compatible with the latest and two immediately preceding major versions of the following web browsers: Google Chrome, Mozilla Firefox, Microsoft	

Requirement ID	Requirement	Description	Remarks
		Edge, and Apple Safari.	

3.2.3 Deployment requirements

Requirement ID	Requirement	Description	Remarks
IMTS-REQ-155	On-Premises	The system shall be installed and operated within the Bank's local infrastructure, ensuring complete control over data security, compliance, and hardware management.	
IMTS-REQ-160	Windows Server OS Platform	The system shall be deployed on Windows Server operating system, ensuring compatibility with enterprise-level features, security protocols, and management tools provided by Microsoft's server platform.	
IMTS-REQ-165	MSSQL Server Database Platform	The system shall be compatible with and utilize Microsoft SQL Server as its primary database management system, ensuring robust data storage, scalability, and integration capabilities within the BOU'S IT infrastructure.	

3.2.4 Security requirements

Requirement ID	Description	Remarks
IMTS-REQ-170	The system shall ensure that the login to the system requires entry of a username and password/passcode.	
IMTS-REQ-175	The system shall not allow concurrent logins for a single user.	

Requirement ID	Description	Remarks
IMTS-REQ-180	The system shall ensure that the password/passcode are masked when the user logs on to the application.	
IMTS-REQ-185	The system shall enable the user to log out of the system.	
IMTS-REQ-190	The system shall not allow successful log in for an invalid user.	
IMTS-REQ-195	The system shall display an error message when a user enters an incorrect username/password combination but not showing whether it is either the username/password.	
IMTS-REQ-200	The system shall be configurable to enforce users to change passwords at first login after account creation or password reset. The system shall require authentication through email at account creation.	
IMTS-REQ-205	The system shall be configurable to allow users to change their passwords as and when necessary. The system shall require authentication through email when changing passwords.	
IMTS-REQ-210	The system shall be configurable to set a maximum password age (in days) that a password can be used before the application requires the user to change it. (Recommended is 90 days)	
IMTS-REQ-215	The system shall be configurable to set a password history such that ten (10) unique new passwords have to be associated with a user account before an old password can be reused.	
IMTS-REQ-220	The system shall be configurable to set password complexity to require at least eight (8) alphanumeric characters of both upper and lower case.	
IMTS-REQ-225	The system shall be configurable to lock an account automatically after three (3) unsuccessful logon attempts.	

Requirement ID	Description	Remarks
IMTS-REQ-230	The system shall be configurable to time out a user's session automatically after ten (10) minutes of inactivity (session time out).	
IMTS-REQ-235	The system shall not disclose user log information after session timeout.	
IMTS-REQ-240	The system shall prior to logging on, specify a legal banner (warning message that user accessing the private computer system, for authorised use only, and activities may be monitored or recorded).	
IMTS-REQ-245	The system shall be accessible over the internet using Secure internet protocols like HTTPS/SSL.	
IMTS-REQ-250	The system shall protect the confidentiality of user transmission between the user client or web browser and the application backend during authentication data transmission (user id, password, passcode).	
IMTS-REQ-255	The system shall record successful user log-in events in an audit log. (These shall be exportable in PDF, Excel, Word or CSV formats).	
IMTS-REQ-260	The system shall record unsuccessful user log-in events in an audit log. (These shall be exportable in PDF, Excel, Word or CSV formats).	
IMTS-REQ-265	The system shall record user log off events in an audit log. (These events shall be exportable in PDF, Excel, Word or CSV formats).	
IMTS-REQ-270	The system shall record all transactions performed by all users in an audit log.	
IMTS-REQ-275	The system shall record account creation events in an audit log.	
IMTS-REQ-280	The system shall record account locking events in an audit	

Requirement ID	Description	Remarks
	log.	
IMTS-REQ-285	The system shall record account disabling events in an audit log.	
IMTS-REQ-290	The system shall record accounts enabling events in an audit log.	
IMTS-REQ-295	The system shall have each audit log entry, at a minimum, containing the following attributes: 1. Log entry identifier 2. Date and time stamp 3. Transaction (event) type 4. User logged on 5. Operation performed 6. Source and destination of IP/Name	
IMTS-REQ-300	The system shall provide the following in a user-friendly way that business users may review: 1. A list of all application users 2. Roles and responsibilities 3. Last access times 4. Logged actions 5. Time stamps for key actions	
IMTS-REQ-305	The system shall only allow users access to their personal dashboards.	
IMTS-REQ-310	The system shall allow assigning responsibilities to users based on roles and groups.	
IMTS-REQ-315	The system shall not allow an unauthorized user to create user accounts and assign user roles.	
IMTS-REQ-320	The system shall allow an authorized user to create user accounts and assign user roles.	
IMTS-REQ-325	The system shall allow user account details to be modified	

Requirement ID	Description	Remarks
	by an authorized user.	
IMTS-REQ-330	The system shall not allow an unauthorized user to modify user account details.	
IMTS-REQ-335	The system shall allow an authorized user to assign and unassign user roles & groups.	
IMTS-REQ-340	The system shall not allow an unauthorized user to assign and unassign user roles & groups.	
IMTS-REQ-345	The system shall allow an authorized user to switch user accounts between the following different states: 1. Active 2. Disabled	
IMTS-REQ-350	The system shall allow end dating of user accounts or roles.	
IMTS-REQ-355	The system shall ensure that access on the system is controlled or restricted to users with appropriate rights.	
IMTS-REQ-360	The system shall allow assignment of users to certain business areas, objectives and initiatives.	
IMTS-REQ-365	The system shall require administrator or super-user privileges on the host system to perform administrative tasks such as start-up/shutdown, general configuration, and troubleshooting.	
IMTS-REQ-370	The system shall function where antivirus software is installed on the systems.	
IMTS-REQ-375	The system shall conform to the respective CIS hardening benchmarks.	
IMTS-REQ-380	The system shall use the latest versions of the client and middle tier software, namely but not limited to: 1. Internet Explorer 11 or later	

Requirement ID	Description	Remarks
	2. Google Chrome version 84 or later 3. Mozilla Firefox version 79.0 or later 4. Java 8 or later 5. Web Server technologies (Apache, OpenSSL, etc.)	
IMTS-REQ-385	The system's application database shall Conform to BOU's database hardening guidelines.	
IMTS-REQ-390	The system shall conform to the respective OWASP hardening benchmarks.	
IMTS-REQ-395	The system shall be installed with the most recently available firmware and software versions, updates and patches.	
IMTS-REQ-400	The system shall not use vulnerable components such as web browsers (Internet Explorer) and Java and Web server libraries, frameworks, and other software modules such as technologies (Apache, OpenSSL, etc.).	
IMTS-REQ-405	The system shall be delivered and configured in accordance with the Bank's policies.	

4 APPENDICES

Appendix I: Flowchart for a Staff User

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Appendix I: Flowchart for a Staff User

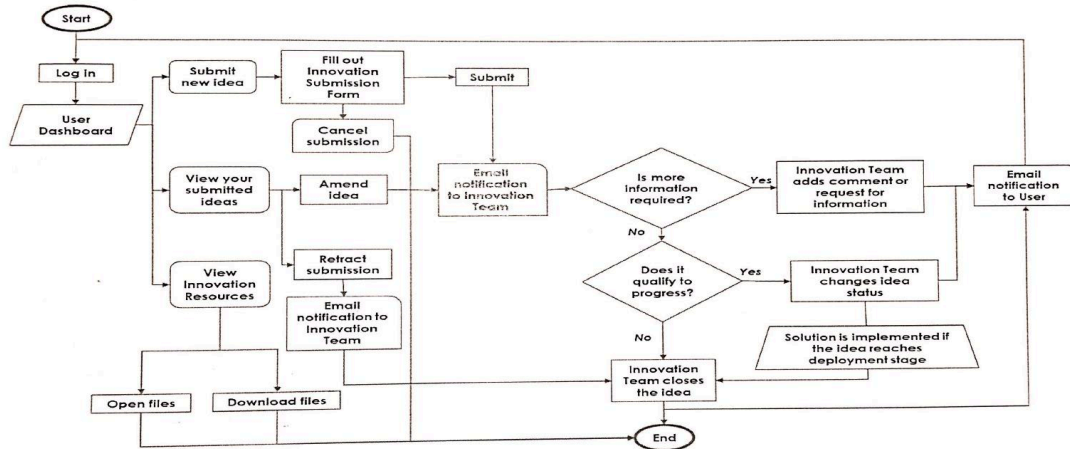


Figure 3: Flowchart for a Staff User

Appendix II: Flowchart for the Innovation Team

[Tviewing

Appendix II: Flowchart for the Innovation Team

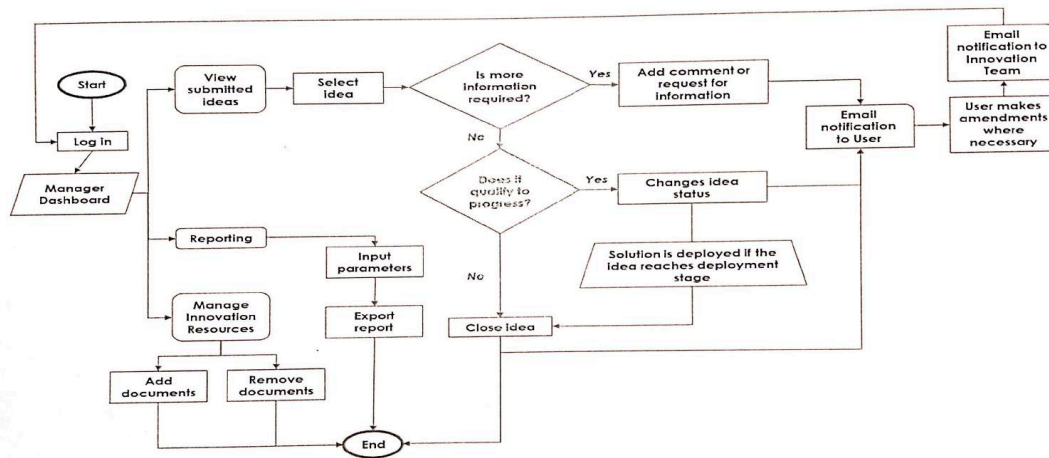


Figure 4: Flowchart for the Innovation Team

Appendix III: Flowchart for the IT Admin

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Appendix III: Flowchart for the IT Admin

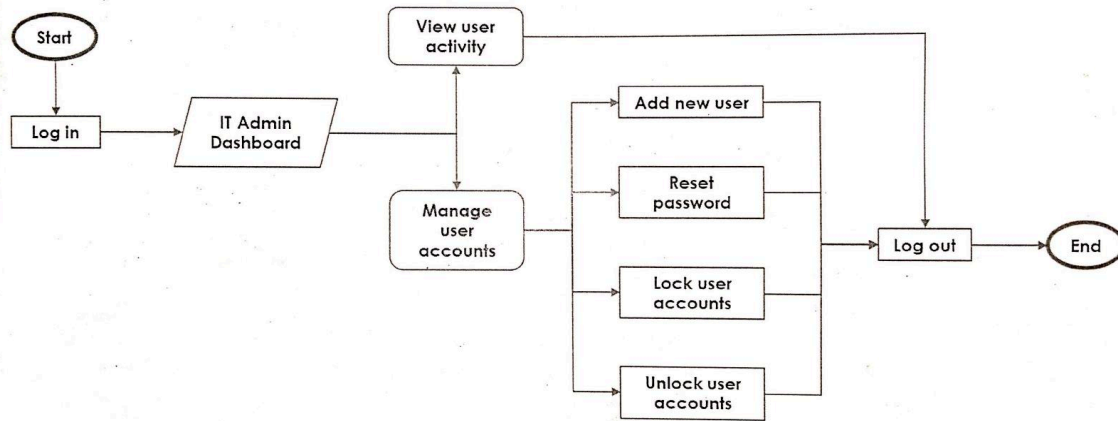


Figure 5: Flowchart for the IT Admin